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SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME:Artificial Intelligence **COURSE CODE: CSA1715**

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 20%

Duration: 1 Hour

QUESTIONS: 5 **Total Marks:50**

Annexure A

Scenario Based Assignment

Code of Conduct:

I _K.Yamini(192572181)_(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

K.Yamini

Signature of the student:

Annexure A

Scenario Based Assignment

1. Scenario : A government deploys AI-powered drones to deliver emergency medicines in a flood-affected region. The environment is highly dynamic—roads may suddenly become inaccessible, and weather conditions affect travel cost. The drone has access to: A heuristic estimate (straight-line distance), Partial real-time updates about blocked paths and Variable movement costs depending on terrain and weather. However, due to urgency, the drone must quickly decide routes with minimum delay and risk.

Tasks:

i. Explain how Greedy Best-First Search would behave in this scenario and discuss its strengths and weaknesses under uncertainty

ii. Explain how A* would handle the same situation, including how it balances cost and heuristic

iii. Analyze the impact of heuristic accuracy on both algorithms

iv. Discuss how dynamic changes (blocked paths) affect search performance

v. Recommend the most suitable algorithm and justify your decision considering real-time constraints, optimality, and safety

2. Scenario: A smart city implements an AI system to optimize traffic signal timings across hundreds of intersections. The objective is to minimize Total waiting time, Fuel consumption and Traffic congestion. The system must continuously adjust signals based on traffic flow. However the search space is extremely large, traffic patterns change dynamically and the system may get stuck in locally optimal solutions

Tasks. Formulate this as an optimization problem and explain why traditional search is inefficient

i. Explain how Hill Climbing would work in this scenario and identify its limitations

ii. Discuss issues like local maxima, plateaus, and ridges with real-world interpretation

iii. Explain how Simulated Annealing or Genetic Algorithms can overcome these limitations

iv. Recommend the best approach and justify your choice based on scalability and adaptability

3. Scenario: An autonomous rover is deployed on Mars to explore unknown terrain. It must: navigate safely to collect samples, avoid hazards (craters, rocks), operate with limited sensing capability and make decisions without a complete map. The rover updates its knowledge as it explores, making this a real-time decision-making problem.

Tasks:

i. Explain why this problem requires an online search agent instead of offline search

ii. Analyze the challenges of partial observability and dynamic environments

iii. Describe how the rover builds and updates its internal model

iv. Compare online search with uninformed and informed search strategies

v. Justify the most suitable approach considering uncertainty, adaptability, and safety

4. Scenario: A university is designing an AI system to automatically generate exam timetables. The system must satisfy multiple constraints: No student should have overlapping exams, Limited number of exam halls, Certain subjects must be scheduled before others, Faculty availability constraints, Special accommodations for some students, The complexity increases as the number of courses and students grows.

Tasks :

Formulate the problem as a Constraint Satisfaction Problem (CSP)

i. Clearly define variables, domains, and constraints with explanation

ii. Explain how backtracking search works in this scenario

iii. Discuss how constraint propagation (e.g., forward checking) improves efficiency

iv. Analyze real-world challenges and justify the best strategy for scalability

5. Scenario: Strategic Game AI for Real-Time Decision Making (Minimax & Alpha-Beta Pruning) A gaming company is developing an AI opponent for a real-time strategy game. The AI must complete against human players, Make decisions within strict time limits, Evaluate

complex game states with multiple possible moves, The decision tree is extremely large, making computation expensive.

Tasks:

i. Explain how the Minimax algorithm models this problem

ii. Analyze the computational challenges of large game trees

iii. Explain how Alpha-Beta pruning improves efficiency with detailed reasoning

iv. Design an evaluation function and justify the factors considered

v. Recommend a strategy for balancing optimality and real-time performance

RUBRICS FOR CALCULATION

Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
		address the scenario			
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

RUBRICS FOR CALCULATION

Criteria	Weightage	Q1 (10)	Q2 (10)	Q3 (10)	Q4 (10)	Q5 (10)	Total (50)
Understanding of Scenario	**20% (2)**	**/2**	**/2**	**/2**	**/2**	**/2**	**/10**
Identification of Key Issues	**20% (2)**	**/2**	**/2**	**/2**	**/2**	**/2**	**/10**
Application of Concepts	**25% (2.5)**	**/2.5**	**/2.5**	**/2.5**	**/2.5**	**/2.5**	**/12.5**
Decision Making	**20% (2)**	**/2**	**/2**	**/2**	**/2**	**/2**	**/10**
Clarity of Response	**15% (1.5)**	**/1.5**	**/1.5**	**/1.5**	**/1.5**	**/1.5**	**/7.5**
Total per Question		**/10**	**/10**	**/10**	**/10**	**/10**	**/50**

Answers

1. Drone Delivery Scenario

i. Greedy Best-First Search Behavior

Greedy Best-First Search selects the path that appears closest to the goal using only the heuristic function (h(n)). In this scenario, the drone will always choose the route with the shortest straight-line distance to the destination without considering actual travel cost.

Strengths:

- Very fast decision-making, suitable for emergency delivery
- Requires less computation
- Works well when heuristic is close to reality

Weaknesses:

- Ignores real-time cost factors such as weather and terrain
- May select dangerous or blocked routes
- Not optimal; may lead to longer or unsafe paths

ii. A Algorithm Handling*

A* uses a combined evaluation function:
$f(n) = g(n) + h(n)$
Where:

- $g(n)$ = actual cost (terrain, weather, obstacles)
- $h(n)$ = heuristic estimate (straight-line distance)

This allows the drone to consider both efficiency and safety while selecting routes.

iii. Impact of Heuristic Accuracy

- If heuristic is accurate → both Greedy and A* perform efficiently
- If heuristic is overestimated → may miss optimal paths
- If heuristic is underestimated → increases computation time

- A* guarantees optimality only when heuristic is admissible

iv. Effect of Dynamic Changes

- Roads may suddenly become blocked due to flooding
- Greedy cannot adapt effectively once path is chosen
- A* can recompute paths when environment changes
- Frequent updates increase computational overhead

v. Recommendation

*A algorithm is most suitable**

- Provides optimal and safe paths
- Balances speed and accuracy
- Handles dynamic updates better than Greedy
- Ensures minimum delay with reduced risk

2. Smart Traffic Signal Optimization

Problem Formulation

- Objective Function: Minimize waiting time, fuel consumption, congestion
- State Space: All possible signal timing combinations
- Constraints: Traffic flow patterns, road capacity, peak hours
- Traditional search is inefficient due to exponential growth of possibilities

i. Hill Climbing Approach

- Starts with an initial solution (signal timing)
- Moves to neighboring solution with better performance
- Stops when no improvement is found

Limitations:

- Cannot escape local optimum
- No backtracking
- Sensitive to initial state

ii. Issues in Real World

- **Local Maxima:** Good solution but not best overall
- **Plateaus:** No improvement in nearby states
- **Ridges:** Path to optimal solution is difficult to find

Example: Traffic signals may stabilize but still cause congestion elsewhere.

iii. Advanced Techniques

Simulated Annealing:

- Allows occasional worse moves
- Helps escape local maxima
- Uses temperature parameter

Genetic Algorithm:

- Uses population of solutions
- Applies selection, crossover, mutation
- Produces better solutions over generations

iv. Recommendation

Genetic Algorithm is best

- Handles large-scale problems efficiently
- Adapts to dynamic traffic conditions
- Provides global optimization
- Suitable for real-time smart city systems

3. Mars Rover Exploration

i. Need for Online Search

- No complete map available initially
- Rover must explore and decide simultaneously
- Cannot precompute full path

ii. Challenges

- Partial observability: limited sensors
- Unknown terrain and hazards
- Dynamic environment changes
- Communication delay with Earth

iii. Internal Model Update

- Stores explored locations
- Updates map with obstacles and terrain features
- Learns safe paths over time
- Uses feedback from sensors

iv. Comparison

- **Uninformed Search:** No knowledge, inefficient
- **Informed Search:** Uses heuristic but needs map
- **Online Search:** Learns and adapts during execution

v. Recommendation

Online Search Agent is most suitable

- Works without full knowledge
- Adapts to new environments
- Ensures safety and exploration efficiency

4. Exam Timetable as CSP

i. CSP Formulation

- **Variables:** Exams (subjects)
- **Domains:** Available time slots
- **Constraints:**
    - No student overlap
    - Limited exam halls
    - Faculty availability
    - Subject ordering requirements
    - Special accommodations

ii. Backtracking Search

- Assigns time slots one by one
- Checks constraints at each step
- Backtracks when violation occurs

iii. Constraint Propagation

- Forward checking removes invalid values early
- Reduces unnecessary exploration
- Improves efficiency significantly

iv. Real-World Challenges

- Large number of students and courses
- Complex and interdependent constraints
- Time and space complexity

Best Strategy:

- Backtracking + Forward Checking + Heuristics (MRV, Degree heuristic)

5. Game AI (Minimax & Alpha-Beta Pruning)

i. Minimax Algorithm

- Models game as a tree of possible moves
- Maximizing player (AI) chooses best move
- Minimizing player (opponent) tries to reduce AI advantage

ii. Computational Challenges

- Exponential growth of game tree
- High memory and time requirements
- Not feasible for real-time without optimization

iii. Alpha-Beta Pruning

- Eliminates branches that do not affect final decision
- Uses alpha (best max value) and beta (best min value)
- Reduces number of nodes evaluated
- Produces same result as Minimax faster

iv. Evaluation Function Design

Factors considered:

- Score or reward
- Number of resources/units
- Position advantage
- Risk level and threats
- Future potential moves

v. Recommendation

Minimax with Alpha-Beta Pruning

- Ensures optimal decision-making

- Reduces computation time
- Use depth limit for real-time performance
- Provides balance between accuracy and speed